

Reconstructing a represented unitary from its shells and residual corner

MathlibAnnex.CStarAlgebra.CAR.exists_targetShellReconstruction

theorem

Separates a represented generator into a strong shell sum and the exact operator between its limiting fixed spaces.

Statement

In the setting below, fix $i \in I$. Put $P_n = \sigma(f_{i,n})$, $Q_n = \sigma(f_n)$ and $W_n = \sigma(w_{i,n})$, and let $F = \bigcap_n \text{ran } P_n$ and $G = \bigcap_n \text{ran } Q_n$. There are contractions S, S^* such that, for every $x \in K$,

$$\sum_{n < N} W_n x \longrightarrow Sx, \quad \sum_{n < N} W_n^* x \longrightarrow S^* x.$$

If P, Q are the orthogonal projections onto F, G , then $S^*S = I - P$ and $SS^* = I - Q$. The remaining operator $R = V_i P$ satisfies

$$V_i = S + R, \quad R^*R = P, \quad RR^* = Q, \quad R = QRP.$$

Moreover, $x \in F$ if and only if $V_i x \in G$, for every $x \in K$.

Assumptions

Let A be the completed CAR algebra, obtained by completing the matrix stages $M_{2^n}(\mathbb{C})$ with embeddings $a \mapsto a \otimes I_2$. Let f_n be the image of the first diagonal matrix unit. Thus $f_0 = 1$ and the f_n form a decreasing sequence of projections. The root state ω has value a_{00} on a matrix a in any stage.

Choose a pure state φ_j in each unitary-equivalence class $j \in I$ of pure-state GNS representations, with $\varphi_o = \omega$ at the root index o . Denote its complete complex GNS space, unital star representation and unit cyclic vector by (H_j, π_j, ξ_j) . Set $\mathcal{H} = \bigoplus_{j \in I} H_j$ and $\pi = \bigoplus_{j \in I} \pi_j$. The index set need not be countable.

A supplied shell family consists of unital complex star automorphisms α_j with $\varphi_j \circ \alpha_j = \omega$ and elements $w_{j,n} \in A$. Put $f_{j,n} = \alpha_j(f_n)$. Their support identities are $w_{j,n}^* w_{j,n} = f_{j,n} - f_{j,n+1}$ and $w_{j,n} w_{j,n}^* = f_n - f_{n+1}$. At the root, α_o is the identity and $w_{o,n} = f_n - f_{n+1}$.

Let L_j be unitaries on $\mathcal{H}$ satisfying $L_j \pi(f_{j,n} - f_{j,n+1}) = \pi(w_{j,n})$ for every j, n . Let $T \subseteq B(\mathcal{H})$ be the norm-closed unital star algebra generated by $\pi(A)$ and all L_j . Write $\iota : A \rightarrow T$ for $\iota(a) = \pi(a)$ and $g_j \in T$ for the element represented by L_j . For a unital complex star representation $\rho : T \rightarrow B(K)$ on a complete complex Hilbert space, put $\sigma = \rho \circ \iota$ and $V_j = \rho(g_j)$. Each V_j is unitary because ρ preserves the unit, products and adjoints. The index i is arbitrary and fixed for this assertion. No irreducibility of ρ , root normalization $L_o = I$, or prescribed action on the selected cyclic vectors is assumed.

Conclusion

The shell sum is a partial isometry from $F^\perp$ onto $G^\perp$, and the residual corner is a partial isometry from F onto G . Together they recover the actual represented generator. The bounds are $\|S\| \leq 1$ and $\|S^*\| \leq 1$; all displayed limits are norm limits for each fixed vector.

The strong sums are formed on K . No continuity of ρ for the strong-operator topology is used, and operator-norm convergence is not asserted. The spaces F, G are not claimed to be one-dimensional in an arbitrary representation. This theorem reconstructs a generator for a supplied family; it does not prove existence of that family.

Proof route

The finite shell identities pass through the star homomorphism and give $V_i(P_n - P_{n+1}) = W_n$. Orthogonality of the initial and final difference projections yields the strong sums of the W_n and their adjoints. Finite telescoping identifies the sum with the complementary part of V_i . Strong convergence of decreasing projections then identifies the remaining corner and its two support projections.

▼ Proof steps

1. Applying ρ to the finite identity $g_i \iota(f_{i,n} - f_{i,n+1}) = \iota(w_{i,n})$ gives the represented shell relation. Also $W_n^* W_n = P_n - P_{n+1}$ and $W_n W_n^* = Q_n - Q_{n+1}$. The two projection sequences start at I and decrease.
2. The difference projections for different n are orthogonal. The orthogonal-shell theorem therefore makes both partial-sum sequences converge strongly to contractions. Taking inner products of the two finite sums identifies their limits as adjoints; the theorem also gives $S^* S = I - P$ and $S S^* = I - Q$.
3. For $s_N = \sum_{n < N} W_n$, telescoping gives $s_N = V_i(I - P_N)$. Its final support identity is $s_N s_N^* = I - Q_N$, hence $V_i P_N V_i^* = Q_N$. Since $P_N \rightarrow P$ and $Q_N \rightarrow Q$ strongly, these finite identities yield $S = V_i(I - P)$ and $V_i P V_i^* = Q$. Only multiplication by fixed bounded operators is used in these limit passages.
4. Set $R = V_i P$. Unitarity and the projection identities give $V_i = S + R$, $R^* R = P$, $R R^* = Q$ and $R = Q R P$. The equality $Q V_i = V_i P$ gives the fixed-space equivalence, in both directions, by injectivity of V_i .

Main citations

- Exact theorem and proof
- The supplied CAR shell-family data
- The norm-closed generated target
- Restriction to the completed CAR source
- The unitary represented generator
- The finite shell identity after representation
- Reconstruction from algebraic source shells
- Telescoping and the limiting residual corner
- The root state bundled with its purity proof

Lean source signature (exact)

```
theorem exists_targetShellReconstruction
  (family : RepresentativeShellFamily)
  (L : MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit →
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
  →L[ $\mathbb{C}$ ]
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState)
  (hLunit :  $\forall i, L i \in \text{unitary}$ 
    (MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState
  →L[ $\mathbb{C}$ ]
    MathlibAnnex.CStarAlgebra.PureState.SelectedAtomicHilbert completedRootPureState))
  (hLsource :  $\forall i n, (L i).comp (selectedAtomicRepresentation
    (transportedFlag family i n - transportedFlag family i (n + 1))) =
    representedShellLink family i n)$ 
  {K : Type v} [NormedAddCommGroup K] [InnerProductSpace  $\mathbb{C}$  K]
  [CompleteSpace K] (rho : Representation (AtomicTarget L) K)
  (i : MathlibAnnex.CStarAlgebra.PureState.GNSClass Limit) :
  let sigma := restrictedRepresentation L rho
  let p := transportedFlag family i
  let q := rootFlag
  let w := (representativeShellData family i).link
  let U :  $\mathbb{N} \rightarrow \text{Submodule } \mathbb{C} K := \text{fun } n \mapsto (\text{sigma } (p n)).\text{range}$ 
  let V :  $\mathbb{N} \rightarrow \text{Submodule } \mathbb{C} K := \text{fun } n \mapsto (\text{sigma } (q n)).\text{range}$ 
   $\exists S T P Q R : K \rightarrow L[\mathbb{C}] K,$ 
  ContinuousLinearMap.StronglyConverges
    (ContinuousLinearMap.partialSum (fun n  $\mapsto$  sigma (w n))) atTop S  $\wedge$ 
  ContinuousLinearMap.StronglyConverges
    (ContinuousLinearMap.partialSum (fun n  $\mapsto$  (sigma (w n)) $^\dagger$ )) atTop T  $\wedge$ 
   $\|S\| \leq 1 \wedge \|T\| \leq 1 \wedge T = S^\dagger \wedge$ 
  IsStarProjection P  $\wedge$  P.range =  $\prod n, U n \wedge$ 
  IsStarProjection Q  $\wedge$  Q.range =  $\prod n, V n \wedge$ 
   $(S^\dagger).comp S = 1 - P \wedge S.comp (S^\dagger) = 1 - Q \wedge$ 
  (Unitary.linearIsometryEquiv
    (representedGeneratorUnitary L hLunit rho i) :  $K \simeq_{L[\mathbb{C}]} K$ ) =
    S + R  $\wedge$ 
  R = ((Unitary.linearIsometryEquiv
    (representedGeneratorUnitary L hLunit rho i) :  $K \simeq_{L[\mathbb{C}]} K$ ) :
     $K \rightarrow L[\mathbb{C}] K$ ).comp P  $\wedge$ 
   $(R^\dagger).comp R = P \wedge R.comp (R^\dagger) = Q \wedge$ 
  R = (Q.comp R).comp P  $\wedge$ 
  ( $\forall x, x \in \prod n, U n \Leftrightarrow$ 
    (Unitary.linearIsometryEquiv
      (representedGeneratorUnitary L hLunit rho i) :  $K \simeq_{L[\mathbb{C}]} K$ ) x  $\in$ 
       $\prod n, V n$ )
```

In the source	Mathematical meaning
<code>family; L; hLunit</code>	The supplied CAR shell family and a family L_i of unitary operators on the selected atomic Hilbert sum $\mathcal{H}$. <code>hLunit</code> states unitarity for every class i .
<code>hLsource : ∀ i n, (L i).comp (...) = representedShellLink family i n</code>	For each i, n , $L_i \pi(f_{i,n} - f_{i,n+1}) = \pi(w_{i,n})$, with $f_{i,n} = \alpha_i(f_n)$. This is the input shell equation; no root normalization or prescribed cyclic-vector action is required of the supplied unitary family. The root conditions within the supplied shell family remain in force.
<code>{K : Type v}; [InnerProductSpace ℂ K]; [CompleteSpace K]</code>	The comparison space K is a complete complex Hilbert space in an arbitrary universe, without a separability requirement.
<code>rho : Representation (AtomicTarget L) K; i : GNSClass Limit</code>	A unital star representation $\rho : T \rightarrow B(K)$ of $T = C^*(\pi(A), \{L_j\})$, and a fixed class i . The representation need not be irreducible.
<code>let sigma := restrictedRepresentation L rho</code>	Set $\sigma = \rho \circ \iota : A \rightarrow B(K)$, where $\iota(a) = \pi(a) \in T$.
<code>let p := transportedFlag family i; let q := rootFlag; let w := (representativeShellData family i).link</code>	These are the algebra-valued sequences $p(n) = f_{i,n}$, $q(n) = f_n$, and $w(n) = w_{i,n}$. Their represented operators are $P_n = \sigma(f_{i,n})$, $Q_n = \sigma(f_n)$, $W_n = \sigma(w_{i,n})$.
<code>let U : ℕ → Submodule ℂ K := fun n ↦ (sigma (p n)).range</code>	The subspaces $U(n) = \text{ran } P_n$ of K , not unitary operators.
<code>let V : ℕ → Submodule ℂ K := fun n ↦ (sigma (q n)).range</code>	The subspaces $V(n) = \text{ran } Q_n$. Their intersection is G ; the intersection of the $U(n)$ is F .
<code>∃ S T P Q R : K →^L[ℂ] K</code>	One simultaneous set of five bounded operators is produced. The witness named <code>T</code> is S^* , not the target algebra T ; the other witnesses are S, P, Q, R .
<code>StronglyConverges (partialSum (fun n ↦ sigma (w n))) atTop S</code>	For each fixed $x \in K$, $\sum_{n < N} W_n x \rightarrow Sx$ in vector norm as $N \rightarrow \infty$.
<code>StronglyConverges (partialSum (fun n ↦ (sigma (w n))[†])) atTop T</code>	For each fixed $x \in K$, $\sum_{n < N} W_n^* x \rightarrow S^* x$ in vector norm. Postfix † denotes the Hilbert adjoint.
<code> S ≤ 1 ∧ T ≤ 1 ∧ T = S[†]</code>	Both limits are contractions and the second is the adjoint of the first: $\ S\ \leq 1$, $\ S^*\ \leq 1$.
<code>IsStarProjection P ∧ P.range = ⋂ n, U n</code>	$P = P^* = P^2$ is the orthogonal projection onto $F = \bigcap_n \text{ran } P_n$.
<code>IsStarProjection Q ∧ Q.range = ⋂ n, V n</code>	$Q = Q^* = Q^2$ is the orthogonal projection onto $G = \bigcap_n \text{ran } Q_n$.

In the source	Mathematical meaning
$(S^\dagger).\text{comp } S = 1 - P \wedge S.\text{comp } (S^\dagger) = 1 - Q$	$S^*S = I - P$ and $SS^* = I - Q$, so S maps $F^\perp$ isometrically onto $G^\perp$.
<code>Unitary.linearIsometryEquiv</code> <code>(representedGeneratorUnitary L hUnit rho i)</code>	The represented generator $V_i = \rho(g_i)$, where $g_i \in T$ is represented by L_i . The wrapper reads this same unitary as a surjective linear isometry, or as a bounded operator when a further cast is displayed.
$\dots = S + R \wedge R = (\dots).\text{comp } P$	The same generator decomposes as $V_i = S + R$, with its residual corner $R = V_iP$.
$(R^\dagger).\text{comp } R = P \wedge R.\text{comp } (R^\dagger) = Q$	The corner has initial support P and final support Q : $R^*R = P$, $RR^* = Q$.
$R = (Q.\text{comp } R).\text{comp } P$	The entire corner is supported from F to G : $R = QRP$.
$\forall x, x \in \prod n, \cup n \leftrightarrow (\dots) x \in \prod n, \forall n$	For every $x \in K$, $x \in F$ if and only if $V_ix \in G$. These are common ranges in the comparison representation, not necessarily one-dimensional. All limits above are strong limits, not operator-norm limits.

Further source notes: Here `Limit` is $\mathcal{A}$, `completedRootPureState` is the root state ω bundled with its purity proof, and `SelectedAtomicHilbert completedRootPureState` is the atomic Hilbert space $\mathcal{H}$. The source witness `T` is the adjoint limit S^* , not the algebra T .

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▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.exists_targetShellReconstruction`

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